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1 Formation of Differential Equations

Before learning the various techniques to solve Differential Equations (DEs), it is important to understand where they come from. A differential equation is typically formed by taking a standard mathematical equation (which represents a family of curves) and eliminating its arbitrary constants.

1.1 1. Concept Overview

Definition: A differential equation is formed by differentiating an ordinary equation and eliminating its arbitrary constants.

When you have an equation representing a family of curves, it will contain one or more unknown parameters, known as **arbitrary constants** (e.g., A, B, C, k). Your goal is to create a new equation that describes the *rate of change* of that family of curves without relying on those specific constants.

1.2 2. The Golden Rule of Formation

To know exactly how many times you need to differentiate the original equation, use this fundamental rule:

The Rule: The number of arbitrary constants in the given general equation is exactly equal to the **Order** of the differential equation you need to form.

- If the equation has **1** arbitrary constant, differentiate **only once**.
- If the equation has **2** arbitrary constants, differentiate **twice**.
- If the equation has n arbitrary constants, differentiate n **times**.

1.3 3. Solved Questions

Here are step-by-step examples from the lecture notes demonstrating how to form differential equations.

1.3.1 Question 1

Form the differential equation for the given equation by eliminating the arbitrary constant. Also, write its order and degree.

$$y = Ax + A^2$$

Solution:

1. **Identify the number of arbitrary constants:** There is only **1** arbitrary constant in this equation, which is A . Therefore, we must differentiate the equation only **once**.
2. **Differentiate with respect to x :**

$$\frac{dy}{dx} = A(1) + 0$$

Let's denote $\frac{dy}{dx}$ as y' . So, $y' = A$.

3. **Eliminate the constant:** Now, substitute the value of A ($A = y'$) back into the original equation $y = Ax + A^2$.

$$y = x(y') + (y')^2$$

Writing this in standard fractional notation:

$$y = x \frac{dy}{dx} + \left(\frac{dy}{dx} \right)^2$$

(This is the required differential equation, free of the constant A .)

4. **Determine Order and Degree:**

- **Order:** The highest derivative is $\frac{dy}{dx}$, so the Order is **1**.
 - **Degree:** The highest power of $\frac{dy}{dx}$ is 2, so the Degree is **2**.
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1.3.2 Question 2

Form the differential equation for the given equation by eliminating the arbitrary constants.

$$y = A \cos x + B \sin x$$

Solution:

1. **Identify the number of arbitrary constants:** There are **2** arbitrary constants in this equation: A and B . Therefore, we must differentiate the equation **twice**.
2. **First Differentiation:** Differentiate y with respect to x :

$$y' = -A \sin x + B \cos x$$

3. **Second Differentiation:** Differentiate y' with respect to x to get the second derivative:

$$y'' = -A \cos x - B \sin x$$

4. **Eliminate the constants:** Notice that in the second derivative, we can factor out a negative sign:

$$y'' = -(A \cos x + B \sin x)$$

Look closely at the expression inside the parentheses; it is exactly our original equation for y ! Substitute y back in:

$$y'' = -y$$

Rearranging to standard form gives us the final differential equation:

$$y'' + y = 0$$

5. **Determine Order and Degree (Bonus):**

- **Order:** The highest derivative is y'' ($\frac{d^2y}{dx^2}$), so the Order is **2**.
 - **Degree:** The power of y'' is 1, so the Degree is **1**.
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Next Steps: You have successfully completed Module 1! You are now ready to begin solving equations. Proceed to [Module 2: Solving First-Order ODEs](#).